

On the robustness of the angular homogeneity scale θ_H : a comparative analysis of computational approaches

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ABSTRACT

Context. The assumption of large-scale homogeneity is a cornerstone of modern cosmology and underlies the validity of the FLRW framework. Testing the scale at which the Universe transitions to homogeneity remains a key observational challenge, particularly with the increasing precision of galaxy surveys.

Aims. We aim to assess the robustness of the angular homogeneity scale, θ_H , by systematically comparing different computational approaches used in its estimation and by quantifying the impact of methodological choices and physical clustering scales.

Methods. We analyse mock galaxy catalogues from the MICE Grand Challenge simulation. The angular fractal dimension $D_2(\theta)$ is computed using the Landy–Szalay estimator and direct pair-counting methods. We implement both parametric and non-parametric approaches, including symbolic regression, to model $D_2(\theta)$ and determine θ_H . Uncertainties are estimated using resampling techniques and alternative propagation methods.

Results. We find that the estimation of θ_H is sensitive to methodological choices in the analysis, including aspects of the numerical implementation and fitting strategy. Differences between approaches can become more pronounced in regimes where θ_H approaches characteristic clustering scales, such as the angular BAO scale. While its redshift evolution is robust, its absolute value is sensitive to both modelling choices and the presence of local clustering features.

Conclusions. Our results highlight the importance of methodological systematics in homogeneity studies, showing that the determination of θ_H depends not only on the data, but also on the adopted analysis strategy. Flexible approaches such as symbolic regression provide a useful framework to model these effects, but also emphasize the need for careful modelling and survey design. This has important implications for future large-scale structure analyses aiming to test the Cosmological Principle with high precision.

Key words. cosmology: theory – cosmology: large-scale structure of Universe – methods: statistical – methods: data analysis – methods: analytical

1. Introduction

One of the foundational assumptions of modern cosmology is the *Cosmological Principle*, the idea that the Universe is statistically homogeneous and isotropic on sufficiently large scales. This principle underpins the standard Λ CDM model based on the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and plays a pivotal role in the interpretation of a wide range of cosmological observations. While the statistical isotropy of the Universe has been generally confirmed, most notably through the analysis of the Cosmic Microwave Background (CMB), direct tests of spatial homogeneity remain more subtle and challenging; see Aluri et al. (2023) and references therein.

To probe homogeneity in the large-scale distribution of matter, statistical tools derived from galaxy clustering are commonly

used. Among these, the fractal dimension D_2 has emerged as a powerful indicator of the scale at which the Universe transitions from a clustered to a homogeneous distribution. When combined with the concept of the three-dimensional homogeneity scale R_H or its angular counterpart θ_H , these measures offer a direct observational way to test the scale of homogeneity across redshifts and assess the validity of the Cosmological Principle.

Historically, the study of cosmic homogeneity began with three-dimensional analyses to determine the spatial fractal dimension $D_2(r)$. Foundational work by Hogg et al. (2005) demonstrated homogeneity using Luminous Red Galaxies, followed by Scrimgeour et al. (2012), which used the WiggleZ Dark Energy Survey to identify a transition to homogeneity scale, R_H , of approximately $70 h^{-1}$ Mpc. Subsequent studies, such as those by Ntelis et al. (2017) and Dias et al. (2023), confirmed R_H to exist within the range of $60 - 80 h^{-1}$ Mpc. However,

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as noted in this work, these 3D measurements are inherently model-dependent, as they require the assumption of a cosmological model to convert observable redshifts into spatial distances. With this mind, our work focuses on the projected (angular) distribution of galaxies in two dimensions and the angular fractal dimension.

The literature has followed two main paths for analyses using the angular fractal dimension: a *direct* method, measuring clustering and D_2 directly from galaxy counts-in-cells, and an *indirect* method, which derives the fractal dimension from the 2PACF. Early applications to surveys such as SDSS and ALFALFA (see Gonçalves et al. (2018); Avila et al. (2018, 2019)) established the standard procedure for identifying the homogeneity scale, θ_H , laying the groundwork for using angular clustering statistics not only to test the transition to homogeneity, but also to explore how this transition may evolve with redshift.

More recent work has improved the statistical reliability and cosmological utility of these measurements. Andrade et al. (2022) introduced a more rigorous approach to uncertainty by using bootstrap resampling of the survey footprint itself, allowing for internal error estimation without relying on external simulations. Most recently, Shao et al. (2025) demonstrated that these measurements are not only consistency checks of the FLRW paradigm; by combining θ_H with other cosmological data, they showed that the transition to homogeneity can be used to constrain cosmological parameters, including dark energy properties, thereby bridging the gap between statistical geometry and fundamental physics.

In parallel with these developments, machine learning techniques have increasingly been explored in cosmology as flexible tools for modelling complex, non-linear phenomena. Among these, symbolic regression (SR) has emerged as a promising approach for reconstructing cosmological functions and providing accurate fitting formulae that can significantly reduce the computational cost of numerically intensive calculations (see e.g. (Arjona & Nesseris 2020; Aizpuru et al. 2021; Nesseris et al. 2022; Carvalho et al. 2023; Carvalho et al. 2026)). Unlike traditional parametric fitting, SR does not assume a predefined functional form, but instead searches the space of mathematical expressions to identify optimal representations of the data.

In this work, we extend the use of symbolic regression to the analysis of fractal dimension profiles $D_2(\theta)$, where, to the best of our knowledge, it has not been previously applied. This approach allows for increased functional flexibility in modelling the $D_2(\theta)$ curves, enabling the capture of local features and small-scale variations associated with galaxy clustering and observational effects. At the same time, it provides analytic expressions that facilitate the determination of the homogeneity scale and its associated uncertainties. This makes SR a useful framework for assessing the robustness of θ_H estimators, particularly in the context of next-generation wide-field galaxy surveys.

Our goal is not only to measure θ_H , but also to quantify the methodological uncertainties associated with its estimation. This is particularly relevant in the context of current and upcoming wide-field galaxy surveys, where percent-level systematics can significantly affect cosmological interpretations. By comparing alternative estimators and computational strategies, we aim to identify robust and reliable procedures for testing the validity of the FLRW paradigm.

This paper is structured as follows. In Section 3, we review the theoretical background of cosmological fractal dimension analysis, the application of this analysis to data and the computational methods proposed to do so. Section 4 describes the

simulated data sets we consider here. In Section 5, we present the results of our different computational approaches. We conclude with a discussion of the implications of our findings in Section 6.

2. Theoretical framework

Within the Standard Model of Cosmology, the assumption of a spatially homogeneous and isotropic Universe leads to the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. Direct probes of the large-scale structure provide an independent data-driven approach to testing this homogeneity.

We adopt a cosmological model-independent approach by examining the projected distribution of objects in the sky, specifically in *two* dimensions. This method utilizes a counts-in-spherical-caps analysis: for a spherical cap of angular radius θ , the number of objects in the observational catalogue is denoted as $N(< \theta)$. For a theoretically homogeneous distribution, the expected count is:

$$N_R(< \theta) = 2\pi\bar{\rho}(1 - \cos \theta) \quad (1)$$

where $\bar{\rho}$ represents the mean number density of pairs of objects in the sky. We define the scaled counts-in-spherical-caps as:

$$N(< \theta) \equiv \frac{N(< \theta)}{N_R(< \theta)} \quad (2)$$

The fractal dimension in two dimensions, $D_2(\theta)$ is expressed as:

$$D_2(\theta) \equiv \frac{d \ln N(< \theta)}{d \ln \theta} = \frac{\theta \sin \theta}{1 - \cos \theta} + \frac{d \ln N(< \theta)}{d \ln \theta} \quad (3)$$

For a distribution that is homogeneous at sufficiently large scales, $D_2(\theta)$ asymptotically approaches D_2^H :

$$D_2^H \equiv \frac{\theta \sin \theta}{1 - \cos \theta} \quad (4)$$

This profile serves as a statistical test of the transverse (angular) homogeneity of the Universe.

2.1. Connection to the 2PACF

Following Peebles (2020), the conditional probability (or differential number counts) of finding a random object within a solid angle element $\delta\Omega$ at an angular separation θ from a specific object is given by:

$$\delta N = \bar{\rho} \delta\Omega [1 + \omega(\theta)], \quad (5)$$

where ω is the two-point angular correlation function (2PACF) and $\bar{\rho}$ is the average density of pairs of objects. By integrating δN over a spherical cap, we obtain the expected number of pairs within radius θ :

$$N(< \theta) = \rho \int d\Omega [1 + \omega(\theta)] \quad (6)$$

This formulation implicitly accounts for counts over the entire survey volume. This leads to the expression for scaled counts:

$$N(< \theta) = 1 + \frac{1}{1 - \cos \theta} \int_0^\theta \omega(\theta') \sin \theta' d\theta' \quad (7)$$

Consequently, the fractal dimension becomes:

$$D_2(\theta) = \frac{\theta \sin \theta}{1 - \cos \theta} + \frac{d}{d \ln \theta} \ln \left[1 + \frac{1}{1 - \cos \theta} \int_0^\theta \omega(\theta') \sin \theta' d\theta' \right] \quad (8)$$

Because the correlation function quantifies matter clustering, its form is heavily dependent on the cosmological model and redshift.

The fractal dimension profile $D_2(\theta)$ allows a clear definition of the angular homogeneity scale: the scale at which $D_2(\theta)$ equals $D_2^H(\theta)$, the homogeneity threshold curve. In practice, this may never happen due to observational artifacts. A commonly used criterion in the literature is the so-called 1% criterion (Scrimgeour et al. 2012; Gonçalves et al. 2017; Avila et al. 2018, 2019; Andrade et al. 2022; Dias et al. 2023; Shao et al. 2025), which defines the homogeneity scale θ_H as the angular radius at which $D_2(\theta)$ first reaches 1% from the homogeneous threshold value:

$$D_2(\theta_H) = 0.99 D_2^H(\theta_H) \quad (9)$$

While this criterion is somewhat arbitrary, it has the advantage of being survey-independent and therefore facilitates comparison across different datasets and against theoretical expected values. Alternative definitions have been proposed in the literature (Hogg et al. 2005; Heinesen 2020), though in this work we adopt the 1% criterion and leave it as an open question whether it is the most adequate to use in future work.

2.2. Numerical estimators

The computation of $\omega(\theta)$, $\mathcal{N}(< \theta)$, and $D_2(\theta)$ from observational data, involves the use of statistical estimators based on pair counts. For a catalogue containing binned redshift data with the right ascension and declination of each object, we determine $DD(\theta)$ by counting object pairs separated by an angular distance θ . We emulate a homogeneous distribution (a random catalogue) by placing objects randomly on the spherical surface considering the original survey footprint, to yield the pair counts $RR(\theta)$. We also account for cross-pairs between the data and the random catalogue, denoted as $DR(\theta)$.

Several estimators for the two-point angular correlation function (2PACF) have been developed from these quantities. Among these, the Landy-Szalay estimator (Eq. (10)) is widely regarded as the most robust (Hirata 2009) and is the choice adopted in this work.

$$\omega_{LS}(\theta) = \frac{DD(\theta) - 2DR(\theta) + RR(\theta)}{RR(\theta)} \quad (10)$$

In all instances, the pair counts DD , DR , and RR are normalized by the total number of pairs in their respective catalogues. This normalization ensures the resulting correlation function is independent of the absolute number of objects, avoiding inherent biases in catalogues of differing sizes.

The scaled counts can be expressed in terms of the Landy-Szalay estimator as:

$$\mathcal{N}_{LS}(< \theta) = 1 + \frac{1}{1 - \cos \theta} \int_0^\theta \omega_{LS}(\theta') \sin \theta' d\theta' \quad (11)$$

Alternatively, we can employ a direct approach by summing pair counts explicitly (see e.g. Fanha 2025):

$$\mathcal{N}_{LS}(< \theta) = 1 + \frac{\sum_{\theta_i=0}^\theta [DD(\theta_i) - 2DR(\theta_i) + RR(\theta_i)]}{\sum_{\theta_i=0}^\theta RR(\theta_i)} \quad (12)$$

3. Methodology

3.1. Computational methodology and uncertainty estimation

While previous works have provided valuable first measurements, they relied on samples with modest galaxy counts and incomplete sky coverage. Next-generation surveys will dramatically improve this landscape. These datasets necessitate a more rigorous statistical framework to determine the angular homogeneity scale, θ_H , and its associated uncertainties.

In general, the analysis follows one of two paths:

- **Indirect Method:** Measuring the 2PACF via pair counts (Eq. 10) and using it to derive the fractal dimension profile $D_2(\theta)$ (Eq. 7 and Eq. 8).
- **Direct Method:** Measuring the fractal dimension curve directly from pair counts (Eq. 12 and Eq. 3).

The scale θ_H is defined as the intersection of the $D_2(\theta)$ curve obtained with a 1% homogeneity threshold.

The indirect approach suffers from one limitation: although obtaining the angular correlation function is more straightforward and one may smooth out the unwanted small-scale oscillations (see e.g. (Avila et al. 2018)), it assumes a perfect, regular footprint for the random distribution, which might not be the case when applying the method to observational data.

We utilize the Landy-Szalay (LS) estimator, with pair counts calculated using the Python package *TreeCorr* at logarithmic-spaced angular separations θ . The number of angular separation bins is determined heuristically to accurately model the $D_2(\theta)$ curve while minimizing oscillations due to the discrete nature of the estimator. For upcoming surveys, the variance introduced by random catalogues is negligible compared to sample variance. To estimate observational uncertainty, we adopt a $\omega(\theta)$ or $D_2(\theta)$ covariance matrix-based approach derived directly from survey data by subdividing the area into spatial patches, taking advantage of the built-in features in *TreeCorr*. We follow the weighting schemes proposed by (Mohammad & Percival 2022) for jackknife and bootstrap resampling. These schemes improve covariance estimation by down-weighting, rather than discarding, information from galaxy pairs crossing patch boundaries. To ensure statistical independence, the typical patch size must exceed the maximum correlation length inferred from $\omega(\theta)$. We determine the ideal number of patches by checking for the convergence of the covariance matrix structure.

Previous studies utilized fixed functional-form fitting, which often fails to reproduce the proper shape of $D_2(\theta)$ across a wide interval of redshift bins. We propose an alternative approach, where we replace these with *Symbolic Regression* (SR), an unsupervised machine learning method that searches the space of mathematical expressions to identify the best fit without predefined parameters. Specifically, given a user-defined list of operators and functions, the building blocks of the mathematical expressions, SR evolves populations of candidate expressions using natural-selection-inspired rules, optimizing both fit accuracy and expression simplicity.

The advantages of SR include:

- **Interpretability:** Produces unbiased/data-driven closed-form expressions with no free parameters that can be compared to theoretical expectations;
- **Flexibility:** Does not require preset functional forms;
- **Uniform Quality:** Guarantees fits across bins via predefined quality thresholds.

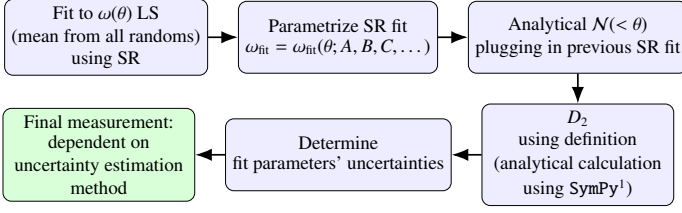


Fig. 1: Flow diagram of the SR approach using the **indirect** method to estimate the angular homogeneity scale and uncertainty.

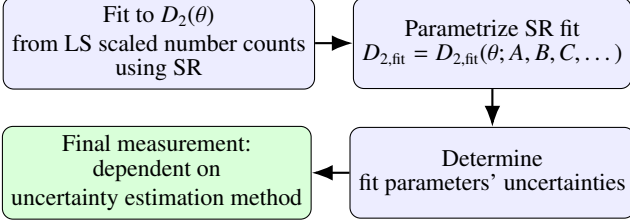


Fig. 2: Flow diagram of the SR approach using the **direct** method to estimate the angular homogeneity scale and uncertainty.

For our direct method, we apply SR to the $D_2(\theta)$ LS data points using basic operators (+, −, ×, ÷, exp) and set complexity at an arbitrary, large value (e.g. 40). The fitness function is defined as the **weighted mean squared error** (WMSE), using the inverse covariance matrix as weights. The stopping criterion is set at $WMSE \leq 1$, meaning residuals are at or below the measurement noise (as such, any other fit that satisfies this condition is equally a good fit and the different expressions found cannot be distinguished). The selection of the best SR candidate expression is determined as a balance of complexity and accuracy (as given by the PySR parameter `model_selection = 'best'`). The two paths, indirect and direct method, can thus be summarized via the two diagrams, Fig. 1 and Fig. 2, respectively.

Uncertainty in θ_H : To estimate the uncertainty of θ_H , we reparametrize the SR expression, replacing constants with variables/parameters while maintaining the mathematical structure, as described in (Fanha 2025), and employ a linear approximation method to estimate the covariance of those parameters relative to the covariance in the data as estimated by the jackknife/resampling techniques mentioned before. The covariance of those parameters is then propagated to θ_H .

Covariance stability and regularization. The cumulative nature of $N(< \theta)$ makes $D_2(\theta)$ highly sensitive to correlation structures, often resulting in covariance matrices with large condition numbers κ . A high κ results in numerical instability, where matrix inversion, used to calculate the WMSE, amplifies errors. To mitigate this, we apply regularization of the covariance matrix C by diagonal loading:

$$C' = C + \lambda I \quad (13)$$

The regularization parameter λ is chosen such that the new covariance matrix C' is structurally similar to C and its smallest eigenvalues differ only by a very small value, enough to stabilize matrix inversion but never result in underestimating uncertainties in θ_H . The value of λ is selected as the maximum value provided by two constraints:

1. **Condition Number Constraint:** Ensuring $\kappa(C) \leq \kappa_{target}$, where:

$$\lambda = \frac{\lambda_{max} - \kappa_{target}\lambda_{min}}{\kappa_{target} - 1} \quad (14)$$

2. **Variance Preservation:** Ensuring λ does not exceed a fraction f of the mean diagonal variance: $\lambda = f \langle \text{diag}(C) \rangle$

Throughout this work, we adopt $\kappa_{target} = 10^4$ and $f = 0.01$ for all calculations.

4. Data

4.1. Galaxy mock Simulations

We use ready-made galaxy mock simulations derived from the Mice Grand Challenge (MICE-GC) N-body simulation, designed to model the large-scale structure of the Universe assuming large-scale homogeneity and isotropy (FLRW cosmology). These simulations provide estimates of galaxy luminosities tailored for wide-field surveys such as the Dark Energy Survey (DES) Abbott et al. (2021) and the Euclid mission Euclid Collaboration: Scaramella et al. (2022). The mock galaxy catalogue is constructed using a hybrid approach that combines a Halo Occupation Distribution (HOD) (see e.g. Jing & Suto (2002); Scoccimarro et al. (2001); Berlind & Weinberg (2002)) model with Halo Abundance Matching (HAM) (see e.g. Vale & Ostriker (2004); Tasitsiomi et al. (2004); Conroy et al. (2006)) to populate Friends-of-Friends (FoF; see e.g. Huchra & Geller (1982); Press & Davis (1982); Tago et al. (1984)) dark matter halos identified in the MICE-GC simulation. The simulation assumes a flat Λ CDM cosmology with matter density parameter $\Omega_m = 0.25$, dark energy density $\Omega_\Lambda = 0.75$, baryon density $\Omega_b = 0.044$, amplitude of matter fluctuations on scales of $8 h^{-1} \text{ Mpc}$ $\sigma_8 = 0.8$, scalar spectral index $n_s = 0.95$, and dimensionless Hubble parameter $h = 0.7$.

The full version 2 of the MICE mock galaxy catalogue Fosalba et al. (2015a); Crocce et al. (2015); Fosalba et al. (2015b) comprises 499,609,997 galaxies, making it one of the largest currently in existence, and is publicly available through the CosmoHub² portal (Carretero et al. 2017). For our analysis, we use a smaller subset of the catalogue, covering roughly 3,438 square degrees from a sky region where DES i-band magnitudes are complete down to $i = 24$, and Euclid H-band magnitudes are complete down to $H \sim 23.0$, up to $z \sim 1.4$, though we select only galaxies up to $i = 23.5$.

We bin the galaxy catalogue in redshift and select bins with left edges at $z = 0.3, 0.5, 0.7, 0.9$, and 1.1 , each with a width of $\Delta z = 0.01$. These redshift bins, together with the adopted magnitude cut and bin width $\Delta z = 0.01$, were chosen so as to ensure good counting statistics in all slices, as indicated in Table 1. The choice of a narrow and fixed redshift width is also motivated by the need to control projection effects in the angular correlation function, which are known to affect the determination of the BAO angular scale (see e.g. Sánchez et al. 2011). In this context, the BAO angular scale is defined as

$$\theta_{\text{BAO}}(z) = \frac{r_s}{(1+z)D_A(z)}, \quad (15)$$

where r_s is the sound horizon at decoupling and $D_A(z)$ is the angular diameter distance at redshift z (see e.g. Aizpuru et al.

² CosmoHub platform: <https://cosmohub.pic.es>

Redshift bin	Average redshift	Number of objects
0.3–0.31	0.305	1,543,290
0.5–0.51	0.505	1,895,610
0.7–0.71	0.705	1,938,871
0.9–0.91	0.905	1,652,408
1.1–1.11	1.105	1,105,749

Table 1: Average redshift and number of objects in each redshift bin for the MICECAT catalogue within the entire completeness region used in this work.

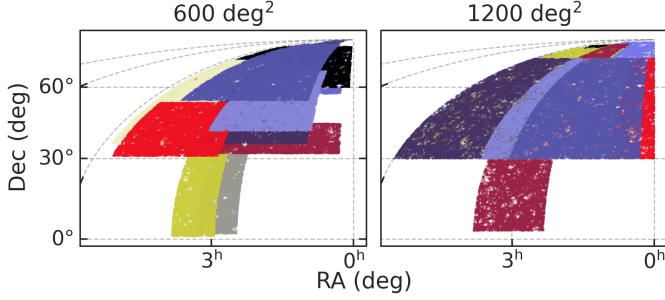


Fig. 3: Distribution on the sky of the 16 regions of 600 deg² (left panel) and of the 16 regions of 1200 deg² (right panel) selected from MICECAT within the completeness region (example for $z = 0.3$). Because of the overlap between regions, many are partially or completely hidden.

2021). For the present choice of redshift slices and for the MICE cosmology adopted here, θ_{BAO} takes values that are close to, and in some cases overlap with, the expected homogeneity scale $\theta_H(z)$ over the redshift range considered. This choice is intentional: it allows us to assess how robust the different estimation methods remain when the angular BAO scale lies near the transition-to-homogeneity scale. More broadly, testing the sensitivity of the methodology under such potentially challenging conditions is one of the goals of this work.

Table 1 presents the average redshift and number of objects in each redshift bin we use in this paper.

4.2. Data analysis procedure

For all analyses, we utilise 30 log-spaced angular separation bins ranging from 0.5° to 20°. To emulate a homogeneous baseline, we generate 20 random catalogues for each survey region, each containing the same number of objects as the data catalogue.

Survey subpartitioning. To analyse how survey area influences θ_H and to assess variability across different sky patches, we subpartition the full catalogue as defined above into multiple, smaller rectangular regions (RA, Dec) with no irregularities or holes. We select (see Fig. 3):

- **16 regions** of 600 deg²;
- **16 regions** of 1200 deg²;
- **One region** covering the full survey area (3400 deg²).

While these subpartitions overlap due to practical limits of the survey footprint, they allow us to mimic the scale of near-future surveys. We note that while differences in region shape could theoretically affect θ_H , our preliminary tests suggest these

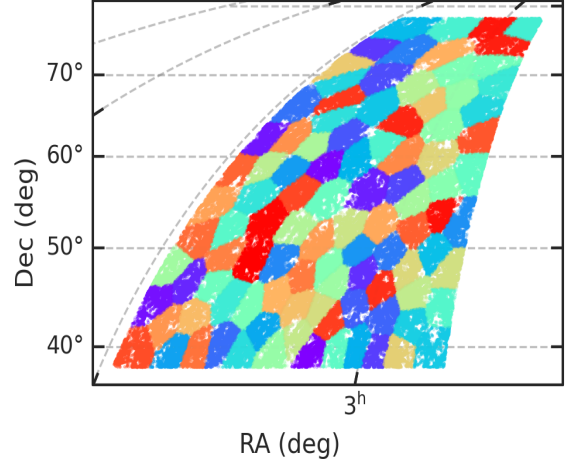


Fig. 4: Example of region subsampled into 100 roughly equal-area patches by TreeCorr for uncertainty estimation.

effects are negligible; however, these findings are not formally presented and should be interpreted with caution.

Patch generation and resampling. To estimate uncertainties on θ_H , we further subdivide each survey region into patches using the *K-means clustering algorithm* built into TreeCorr. This algorithm produces patches of roughly equal area.

The number of patches is subject to two competing constraints:

1. **Lower limit:** To ensure enough statistical samples of $\omega(\theta)$ and $D_2(\theta)$, jackknife/bootstrap resampling needs at least as many patches as angular separation bins;
2. **Upper limit:** Each patch must be large enough so its typical scale exceeds the correlation length of the region, ensuring the statistical independence required for resampling.

We vary the number of patches up to **200** for the full-area region, staying safely below the upper limit while maximizing the resolution of within-region variability. In Figure 4 we plot an example of the patch divided into 100 equal areas. Our observations indicate:

- **Full-Area region:** The covariance structure stabilizes once $N_{\text{patch}} \geq 100$. To ensure maximum robustness while remaining below the physical correlation limit, we adopt $N_{\text{patch}} = 200$ for this region;
- **1200 deg² regions:** Stability is reached at approximately 100-150 patches;
- **600 deg² regions:** We selected 50 patches to maintain a typical scale in a patch smaller than the correlation length of the survey.

We compared bootstrap and jackknife resampling. While both methods estimate the covariance matrix with sufficient accuracy, we adopt jackknife resampling as the preferred method for all subsequent analyses due to being:

- **more computationally efficient:** jackknife is significantly faster as the number of iterations always equals the number of patches, whereas bootstrap requires a much higher number of iterations to converge;
- **physically consistent:** jackknife does not generate physically unrealistic resampled regions.

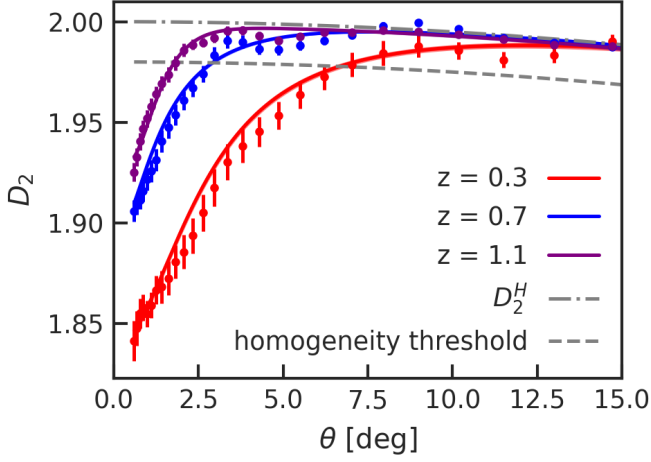


Fig. 5: Representative symbolic-regression reconstructions of the profiles $D_2(\theta)$ for three redshift bins in the full-area MICECAT region. Points with error bars correspond to the measured profiles, solid curves show the symbolic-regression fits, and the horizontal threshold marks the homogeneity criterion.

Selection of the method for calculating D_2 . Following our comparative testing of both analytical paths, we have opted to utilize the direct method for the remainder of this work. This choice was motivated by the fact that its implementation is simpler and less prone to numerical instability and that it does not suffer from the limitation described in Section 3 regarding the theoretical prediction for random counts.

5. Results

5.1. Angular correlation and $D_2(\theta)$ determination

Fig. 5 shows three representative profiles of $D_2(\theta)$ obtained from the direct estimator together with their symbolic-regression reconstructions. For clarity, only the redshift bins $z = 0.3$, $z = 0.7$, and $z = 1.1$ are displayed, spanning the low-, intermediate-, and high-redshift regimes probed in this work.

The corresponding analytical expressions for all redshift bins are listed in Table 2, where we report the reconstructed forms for $D_2^*(\theta) = D_2(\theta) - 0.99D_2^H(\theta)$ and their mean squared errors. The symbolic-regression models were selected using the `model_selection='best'` criterion in PySR, which identifies the expression lying closest to the optimal trade-off between accuracy and complexity along the Pareto front. This selection avoids overfitting while retaining sufficient flexibility to capture the relevant features of the data.

At small angular scales, $D_2(\theta)$ departs significantly from homogeneity, reflecting the clustered nature of the galaxy distribution, while at larger angular separations it gradually approaches the homogeneous limit $D_2^H(\theta)$ given by Eq. (4). Superimposed on this global trend, the observed profiles exhibit local fluctuations, which may arise from genuine structure in the matter distribution, such as collapsed systems and voids, but also from survey-related effects, including incomplete sky coverage and the imprint of BAO-scale clustering.

In this context, symbolic regression offers an important advantage over standard parametric fitting functions. Rather than imposing a fixed functional form, it allows the data themselves to determine a flexible analytic representation of the profile. This added flexibility enables the fit to absorb fluctuations in strongly

Redshift	MSE	Expression for $D_2^*(\theta)$	Parameters
$z = 0.3$	4.85×10^{-5}	$a_0 + \frac{a_1}{a_2 + a_3\theta^2}$	$a_0 = 0.0235$ $a_1 = 0.3260$ $a_2 = -1.9807$ $a_3 = -0.2673$
$z = 0.5$	7.01×10^{-5}	$a_0 + a_1\theta a_2^\theta$	$a_0 = 0.0185$ $a_1 = -0.3746$ $a_2 = 0.2876$
$z = 0.7$	2.30×10^{-5}	$a_0 + \frac{a_1^\theta}{a_2 + a_3\theta + a_4 \exp(\exp(a_5\theta))}$	$a_0 = 0.0194$ $a_1 = 0.6931$ $a_2 = -0.8000$ $a_3 = -4.5651$ $a_4 = -3.5651$ $a_5 = -1.4601$
$z = 0.9$	4.65×10^{-5}	$a_0 + a_1 \exp(a_2\theta)$	$a_0 = 0.0203$ $a_1 = -0.1139$ $a_2 = -0.6402$
$z = 1.1$	7.14×10^{-6}	$a_0 + a_1\theta \exp(-2\theta)$	$a_0 = 0.0179$ $a_1 = -0.3599$

Table 2: Mean squared error and symbolic regression expressions for $D_2^*(\theta) = D_2(\theta) - 0.99D_2^H(\theta)$ in different redshift bins.

inhomogeneous regimes ($\theta < \theta_H$) and in scales already approaching homogeneity ($\theta \gtrsim \theta_H$), while preserving a stable description of the transition region itself. As a result, the inferred intersection with the homogeneity threshold is less sensitive to local distortions of the $D_2(\theta)$ curve than in conventional global parametric fits, and avoids the limitations of local fits restricted to the neighbourhood of the intersection, which can themselves be strongly affected by fluctuations at those angular scales.

The reconstructed expressions listed in Table 2 also show that the symbolic-regression solutions remain relatively compact, while adapting their complexity to the structure of each redshift bin. In particular, the higher-redshift profiles are captured by simpler saturating forms, whereas the lower-redshift cases require more structured expressions, consistent with the stronger clustering and increased profile irregularity expected at later cosmic times. These functions should not be interpreted as fundamental physical laws, but rather as effective analytic representations of $D_2^*(\theta)$ for the MICECAT simulation and cosmology adopted here.

5.2. Uncertainty estimation

A key aspect of the determination of the homogeneity scale θ_H is the robust estimation of its associated uncertainty. In this work, we compare three different approaches (an in-depth description can be found in Fanha (2025)):

- A linear approximation based on error propagation,
- A nested sampling method applied to the symbolic regression (SR) parametrisation,
- And a third, parametric, resampling-based approach, inspired by existing methods in the literature, where:
 1. From jackknife resamples of the survey region, we generate an array of D_2 profiles, describing the intrinsic variability of D_2 within that region,
 2. We generate the distribution of θ_H values for this region by fitting a spline over each of the D_2 profiles and interpolating to find the angular homogeneity scale,
 3. We estimate the most likely θ_H value and the uncertainty interval as the 16/84% percentiles of this distribution.

The resulting 1σ uncertainties for the full-area region are shown in Fig. 6. We find that the linear approximation and nested

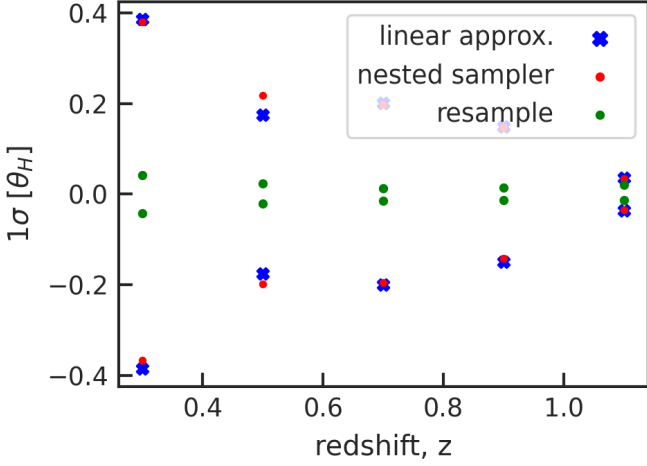


Fig. 6: Uncertainties estimated on θ_H measurements for the full area region using different methods - linear approximation and nested sampler (from SR approach, described in Section 3) and resample, applying the direct method for θ_H estimation. The negative and positive values correspond to the lower and upper bounds of the standard deviation.

sampling approaches yield broadly consistent results, indicating that both methods capture similar features of the underlying parameter space. However, both tend to produce larger uncertainty estimates when compared to the resampling-based method.

This behaviour reflects fundamental differences between the approaches. The linear approximation and nested sampling methods rely on the parametrised form of the $D_2(\theta)$ profile and propagate uncertainties through the fitted model, making them sensitive to the assumed functional representation. In contrast, the resampling-based approach directly probes the variability of the data by accounting for spatial fluctuations within the survey footprint. Besides these differences, the θ_H estimated via each method are also necessarily different: in the resampling approach, it would not be computationally feasible to produce SR profiles for each D_2 resample, so we fit linear splines, which make the θ_H estimates more vulnerable to small-scale oscillations from effects previously enumerated. In the nested sampling approach, the θ_H estimates are also determined by a distribution of θ_H resulting from the nested sampler samples. Ultimately, the linear approximation method is simpler and more straightforward to implement, while not producing estimates that are too conservative, in comparison.

5.3. Survey area dependence

We investigate the impact of survey area on the stability of the homogeneity scale θ_H by analysing multiple realizations of regions with areas of 600 deg^2 (top panels) and 1200 deg^2 (bottom panels), for both the non-parametric (SR) and parametric resampling approaches (described in 5.2 and in Fanha (2025)), as shown in Fig. 7 and Fig. 8.

In both figures, in each panel, each point represents an individual measurement of θ_H obtained from a given region realization (see Fig. 4), while the associated error bars correspond to the uncertainties estimated via the linear approximation approach. The vertical dashed lines indicate the mean θ_H and the corresponding $\pm 1\sigma$ intervals inferred from bootstrapping over the sixteen region realizations within the full-area region for each red-

shift bin. The distributions shown in each panel thus provide a direct visualization of the dispersion in θ_H across different regions for a given survey area. A comparison with the corresponding full-area measurements is presented in the following section.

A clear dependence on survey area is observed in both approaches. The 600 deg^2 regions exhibit a significantly larger scatter in θ_H values compared to the 1200 deg^2 regions, reflecting the stronger impact of sample variance in smaller survey volumes. As the survey area increases, the distributions become more concentrated and the separation between redshift bins becomes more pronounced, indicating improved stability of the inferred homogeneity scale.

However, important differences emerge when comparing the two fitting methodologies. The non-parametric (SR) approach shows a significantly larger dispersion in θ_H across subregions, together with broader and more structured distributions. In contrast, the parametric approach yields more compact and regular distributions, indicating a reduced sensitivity to local fluctuations in the $D_2(\theta)$ profile. In addition, we find a bias in the θ_H estimates between the two approaches.

These differences become particularly evident in specific redshift bins. For example, at $z = 0.9$, the SR-based estimates in the 600 deg^2 regions show a noticeable departure from the expected monotonic decrease of θ_H with redshift (see the top panel of Fig 7). This effect is less pronounced for larger areas and is largely absent in the parametric case, where the expected ordering of θ_H with redshift is preserved.

This behaviour can be understood in light of the proximity between the homogeneity scale and the angular BAO scale. As discussed in the following section, for the adopted redshift binning and selection criteria, θ_H approaches θ_{BAO} at specific redshifts. In this regime, local features in the $D_2(\theta)$ profile induced by BAO-scale clustering can significantly affect the determination of the transition scale, particularly in smaller survey areas where such features are less efficiently averaged out.

Because the SR method allows for a more flexible representation of the data, it is more sensitive to these local distortions, resulting in increased variability across subregions. In contrast, the parametric approach enforces a predefined smooth functional form, which tends to suppress such fluctuations and yields more stable, but potentially biased, estimates of θ_H .

It is also important to note that, due to the finite size of the simulated catalogue ($\sim 3400 \text{ deg}^2$), the different realizations of subregions are not fully independent and exhibit partial overlap. This overlap is random between realizations but implies that the sampled volumes are not statistically disjoint. As a result, the observed dispersion likely underestimates the true cosmic variance expected for fully independent survey regions of the same size, representing an additional limitation associated with the finite survey area.

Overall, these results highlight the interplay between survey geometry, physical clustering scales, and methodological choices. While larger survey areas lead to more robust estimates of θ_H , the comparison between fitting approaches reveals that the inferred homogeneity scale can be sensitive to the presence of BAO-scale features, particularly when analysing smaller regions. This reinforces the need to carefully assess methodological systematics when interpreting homogeneity measurements in current and future large-scale surveys.

5.4. Redshift evolution of the homogeneity scale

Finally, we present the measurements of the homogeneity scale, expressed as θ_H , as a function of redshift for both the non-

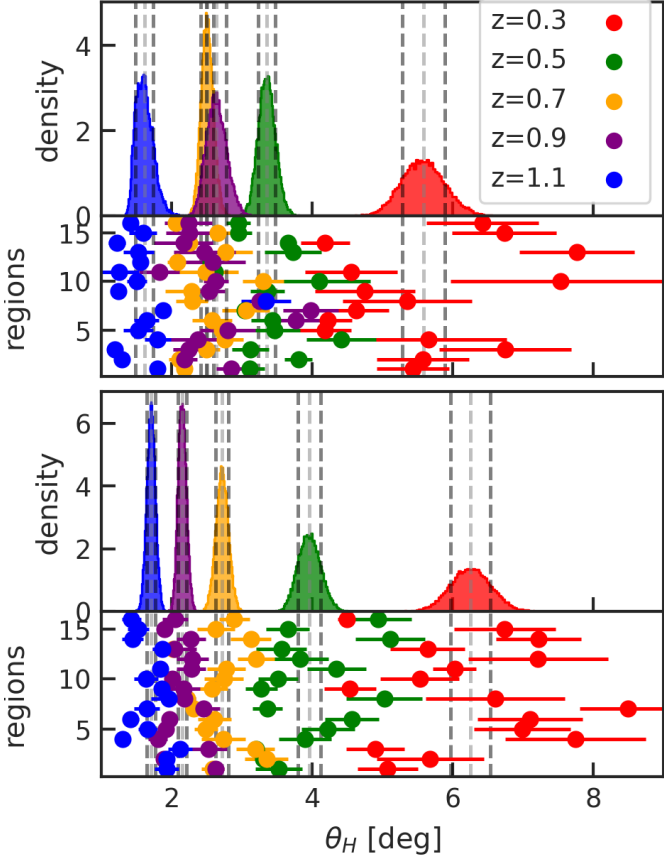


Fig. 7: **Top:** Results for regions of 1200 deg². **Bottom:** Results for regions of 600 deg².

Histograms of bootstrap distributions from θ_H measurements (SR) plus uncertainties via the linear approximation approach with the direct method over many different regions of different areas.

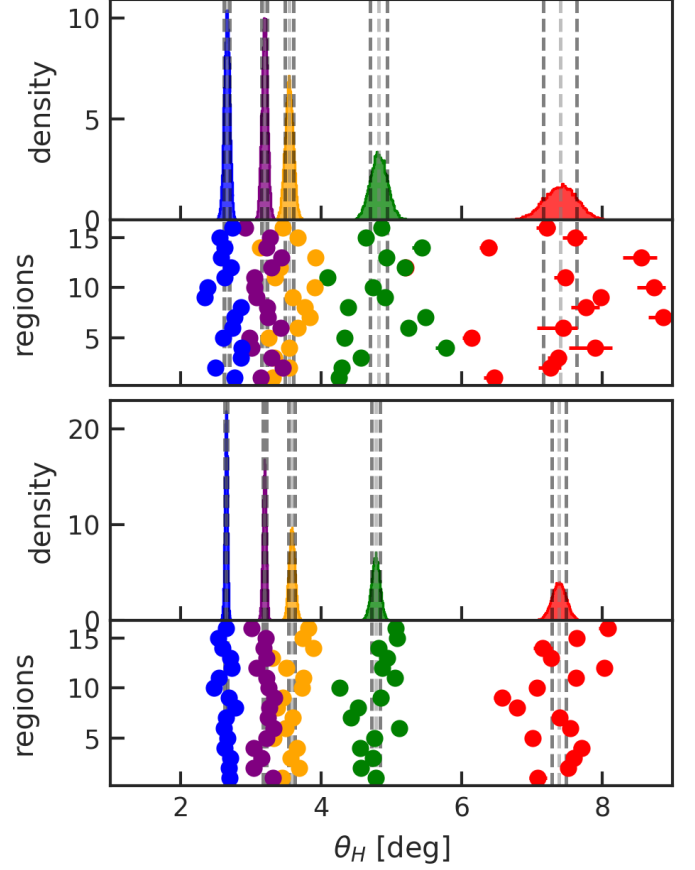


Fig. 8: **Top:** Results for regions of 1200 deg². **Bottom:** Results for regions of 600 deg².

Histograms of bootstrap distributions from θ_H measurements (resampling) plus uncertainties with the direct method over many different regions of different areas.

parametric (SR) and parametric resampling approaches, shown in Fig. 9.

A clear and systematic decrease of θ_H with increasing redshift is observed in both approaches. This behaviour is expected, as a fixed physical homogeneity scale subtends a smaller angular size at higher redshift due to the evolution of the angular diameter distance. The overall trend is therefore consistent with expectations from standard cosmological models.

For both methods, the comparison between different survey areas shows that the mean values obtained from 600 deg² and 1200 deg² regions tend to converge towards the full-area measurement for $z \gtrsim 0.5$. This confirms that, despite the increased dispersion observed in smaller regions, averaging over multiple subregions provides a reliable estimate of the global homogeneity scale.

However, a significant difference between the two approaches is observed in the absolute values of θ_H . The non-parametric (SR) method systematically yields lower values of θ_H compared to the parametric approach, with discrepancies that exceed the corresponding uncertainty estimates in several redshift bins. This indicates that the determination of θ_H is not only sensitive to survey area, but also to the adopted fitting methodology.

This behaviour can be understood by considering the proximity between the homogeneity scale and the angular BAO scale. As shown in Fig. 9, the inferred values of θ_H lie close to $\theta_{BAO}(z)$ over a significant fraction of the redshift range. In this regime, local distortions of the $D_2(\theta)$ profile associated with BAO-scale structure can affect the determination of the intersection with the homogeneity threshold.

Because the two methods adopt fundamentally different fitting strategies, they respond differently to such features. The parametric approach enforces a predefined functional form, which tends to smooth over local fluctuations, while the symbolic regression method allows the data to determine a more flexible representation. As a result, the inferred transition scale becomes method dependent when θ_H lies close to the BAO angular scale.

This interpretation is supported by the observed systematic offset between the two methods and by the increased sensitivity of the SR-based estimates to variations across survey regions. It suggests that, in regimes where $\theta_H \sim \theta_{BAO}$, the homogeneity scale should be interpreted with caution, as it may be influenced by the presence of BAO-induced structure in the underlying clustering signal.

Overall, these results demonstrate that while the redshift evolution of θ_H is robust, its absolute determination can be affected

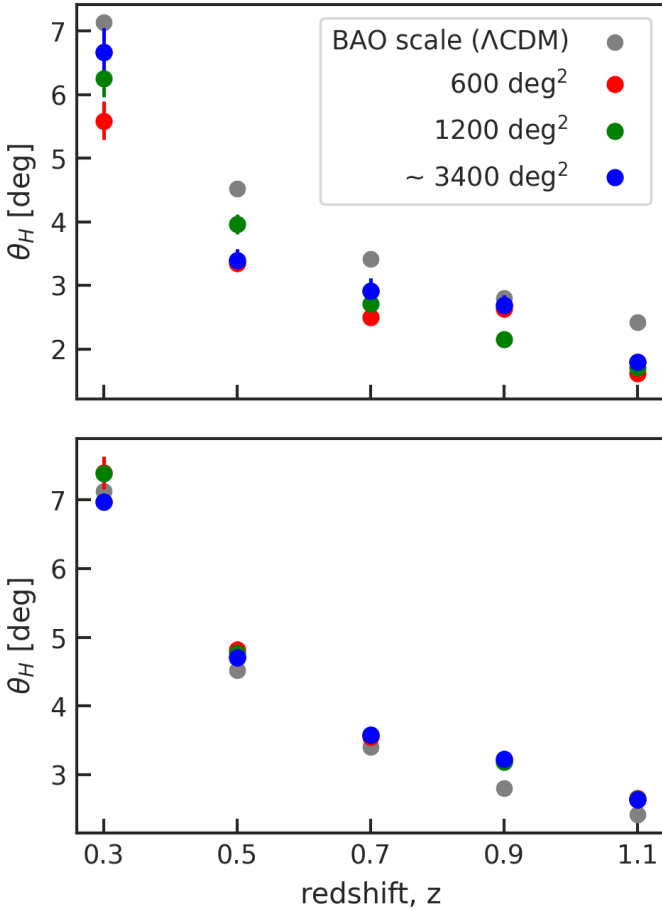


Fig. 9: **Top:** Results for non-parametric SR approach. **Bottom:** Results for parametric resampling approach. Evolution of the homogeneity scale θ_H as a function of redshift. Measurements from subregions of 600 deg^2 (red) and 1200 deg^2 (green) are compared with the estimate from the full-area region ($\sim 3400 \text{ deg}^2$, blue). The grey points indicate the expected BAO angular scale computed from a fiducial ΛCDM cosmology.

by both methodological choices and the proximity to other physical scales. This highlights the importance of jointly considering modelling assumptions and survey characteristics when using θ_H as a cosmological probe.

6. Conclusions

In this work, we have presented a systematic assessment of different computational approaches to estimate the angular homogeneity scale, θ_H , using simulated galaxy catalogues from the MICE Grand Challenge. Our goal was to evaluate the robustness of θ_H measurements and to quantify the impact of methodological choices, survey geometry, and physical clustering features on its determination.

We first analysed the reconstruction of the fractal dimension profile $D_2(\theta)$ using a non-parametric symbolic regression (SR) approach applied to the direct estimator. This method provides flexible analytic representations of the data without imposing predefined functional forms, allowing the reconstructed profiles to adapt to both the strongly inhomogeneous regime and the transition to homogeneity. The resulting expressions remain

compact while capturing the relevant features of the measured profiles across different redshifts.

We then investigated different strategies for estimating the uncertainty on θ_H . We found that approaches based on linear error propagation and nested sampling yield consistent results, although they tend to produce more conservative uncertainties compared to resampling-based methods. The latter directly probe the variability of the data within the survey footprint and are therefore better suited to capture the impact of sample variance and survey geometry.

The dependence of θ_H on survey area was examined by analysing multiple realizations of subregions. As expected, smaller areas exhibit significantly larger dispersion due to increased sensitivity to local inhomogeneities and large-scale modes, while larger areas provide more stable and reproducible estimates. Nevertheless, averaging over multiple subregions yields mean values consistent with those obtained from the full survey area, indicating that reliable global estimates can be recovered even from smaller patches when treated statistically.

A clear and systematic decrease of θ_H with redshift was observed, consistent with the expected scaling of angular sizes with cosmic distance. This behaviour is robust across different survey areas and estimation methods, supporting the use of θ_H as a tracer of large-scale structure evolution.

However, a key result of this work is the identification of a significant dependence of the inferred homogeneity scale on the adopted fitting methodology. The non-parametric (SR-based) and parametric approaches yield systematically different values of θ_H , with discrepancies that exceed the corresponding uncertainty estimates in several redshift bins. This demonstrates that the determination of θ_H is not purely data-driven, but is influenced by the modelling assumptions used to describe the $D_2(\theta)$ profile.

We interpret this behaviour in the context of the proximity between the homogeneity scale and the angular BAO scale. For the redshift binning and selection criteria adopted in this work, θ_H approaches θ_{BAO} in several cases. In this regime, local features in the clustering signal associated with BAO can affect the determination of the intersection with the homogeneity threshold. Because parametric and non-parametric approaches respond differently to such features, the inferred value of θ_H becomes method dependent. While the impact of BAO on homogeneity measurements has been discussed in previous works, our results highlight how this effect can manifest differently depending on the adopted reconstruction strategy.

From a methodological perspective, symbolic regression provides a flexible alternative to traditional parametric fits, allowing the data to determine the functional form of $D_2(\theta)$. This flexibility enhances the ability to capture local features in the profile, but also makes the resulting θ_H estimates more sensitive to fluctuations, particularly in smaller survey areas or in regimes where physical scales overlap. In contrast, parametric models tend to smooth such features, yielding more stable but potentially biased estimates.

Overall, our results show that while the redshift evolution and large-scale behaviour of θ_H are robust, its absolute determination can be sensitive to both methodological choices and the presence of nearby physical scales such as the BAO feature. This highlights the importance of carefully designing analysis strategies, including redshift binning and sample selection, to minimize such effects when using θ_H as a cosmological probe.

Looking ahead, future large-scale surveys will provide significantly improved statistics and sky coverage, enabling more precise measurements of the homogeneity scale. In this context,

approaches such as symbolic regression may offer a valuable tool for flexible modelling, while systematic comparisons between methods will remain essential to ensure the robustness of cosmological inferences.

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References

- Abbott, T. M. C., Adamów, M., Agüena, M., et al. 2021, *Astrophysical Journal*, Supplement Series, 255, 20
- Aizpuru, A., Arjona, R., & Nesseris, S. 2021, *Phys. Rev. D*, 104, 043521
- Aluri, P. k., Cea, P., Chingangbam, P., et al. 2023, *Classical and Quantum Gravity*, 40, 094001
- Andrade, U., Gonçalves, R. S., Carvalho, G. C., et al. 2022, The angular scale of homogeneity with SDSS-IV DR16 Luminous Red Galaxies
- Arjona, R. & Nesseris, S. 2020, *Phys. Rev. D*, 101, 123525
- Avila, F., Novaes, C. P., Bernui, A., & de Carvalho, E. 2018, *Journal of Cosmology and Astroparticle Physics*, 2018, 041–041
- Avila, F., Novaes, C. P., Bernui, A., de Carvalho, E., & Nogueira-Cavalcante, J. P. 2019, *Monthly Notices of the Royal Astronomical Society*, 488, 1481–1487
- Berlind, A. A. & Weinberg, D. H. 2002, *The Astrophysical Journal*, 575, 587
- Carretero, J., Tallada, P., Casals, J., et al. 2017, in *Proceedings of the European Physical Society Conference on High Energy Physics*. 5–12 July, 488
- Carvalho, A., Krone-Martins, A., Da Silva, A., Mimoso, J. P., & Böhm, C. 2026, arXiv e-prints, arXiv:2603.21277
- Carvalho, A., Oliveira, D. M., Krone-Martins, A., & Da Silva, A. 2023, in *2023 IEEE 19th International Conference on e-Science (e-Science)*, 1–2
- Conroy, C., Wechsler, R. H., & Kravtsov, A. V. 2006, *The Astrophysical Journal*, 647, 201
- Crocce, M., Castander, F. J., Gaztañaga, E., Fosalba, P., & Carretero, J. 2015, *Monthly Notices of the Royal Astronomical Society*, 453, 1513
- Dias, B. L., Avila, F., & Bernui, A. 2023, Probing cosmic homogeneity in the Local Universe
- Euclid Collaboration: Scaramella, R., Amiaux, J., Mellier, Y., et al. 2022, *Astronomy and Astrophysics*, 662, A112
- Fanha, P. M. S. 2025, Master's dissertation, University of Porto, Porto, Portugal, accessed: 2026-03-28
- Fosalba, P., Crocce, M., Gaztañaga, E., & Castander, F. J. 2015a, *Monthly Notices of the Royal Astronomical Society*, 448, 2987
- Fosalba, P., Gaztañaga, E., Castander, F. J., & Crocce, M. 2015b, *Monthly Notices of the Royal Astronomical Society*, 447, 1319
- Gonçalves, R. S., Carvalho, G. C., Bengaly, Jr., C. A. P., et al. 2018, *MNRAS*, 475, L20
- Gonçalves, R. S., Carvalho, G. C., Bengaly Jr., C. A. P., et al. 2017, *Monthly Notices of the Royal Astronomical Society: Letters*, 475, L20–L24
- Heinesen, A. 2020, *Journal of Cosmology and Astroparticle Physics*, 2020, 052–052
- Hirata, C. M. 2009, Correlation functions: Generalities, draft: November 19, 2009, Caltech cosmology notes.
- Hogg, D. W., Eisenstein, D. J., Blanton, M. R., et al. 2005, *The Astrophysical Journal*, 624, 54–58
- Huchra, J. P. & Geller, M. J. 1982, *The Astrophysical Journal*, 257, 423
- Jing, Y. P. & Suto, Y. 2002, *The Astrophysical Journal*, 574, 538–553
- Mohammad, F. G. & Percival, W. J. 2022, *Monthly Notices of the Royal Astronomical Society*, 514, 1289–1301
- Nesseris, S., Sapone, D., Martinelli, M., et al. 2022, *A&A*, 660, A67
- Ntelis, P., Hamilton, J.-C., Goff, J.-M. L., et al. 2017, *Journal of Cosmology and Astroparticle Physics*, 2017, 019–019
- Peebles, P. J. E. 2020, *Principles of Physical Cosmology* (Princeton University Press)
- Press, W. H. & Davis, M. 1982, *The Astrophysical Journal*, 259, 449
- Scoccimarro, R., Sheth, R. K., Hui, L., & Jain, B. 2001, *The Astrophysical Journal*, 546, 20
- Scrimgeour, M. I., Davis, T., Blake, C., et al. 2012, *Monthly Notices of the Royal Astronomical Society*, 425, 116–134
- Shao, X., Bengaly, C. A. P., Gonçalves, R. S., Carvalho, G. C., & Alcaniz, J. 2025, *European Physical Journal C*, 85, 225
- Shao, X., Bengaly, C. A. P., Gonçalves, R. S., Carvalho, G. C., & Alcaniz, J. 2025, Cosmological constraints from angular homogeneity scale measurements
- Sánchez, E., Carnero, A., García-Bellido, J., et al. 2011, *Monthly Notices of the Royal Astronomical Society*, 411, 277
- Tago, E., Einasto, J., & Saar, E. 1984, *Monthly Notices of the Royal Astronomical Society*, 206, 559
- Tsitsiomi, A., Kravtsov, A. V., Wechsler, R. H., & Primack, J. R. 2004, *The Astrophysical Journal*, 614, 533
- Vale, A. & Ostriker, J. P. 2004, *Monthly Notices of the Royal Astronomical Society*, 353, 189